1. Create an index that will enable a user to pull orders for ‘1990-10-03’ out of the Orders table quickly.

```
d5_96305_sanchit> select * from orders where odate = '1990-10-03';
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
5 rows in set (0.157 sec)

d5_96305_sanchit> explain analyze select * from orders where odate = '1990-10-03';
+-----------------------------------------------------------------------------------------------------------------------------------
----------+
| EXPLAIN
          |
+-----------------------------------------------------------------------------------------------------------------------------------
----------+
| -> Index lookup on orders using i_date (odate = DATE'1990-10-03')  (cost=1 rows=5) (actual time=0.0272..0.0376 rows=5
loops=1)
 |
+-----------------------------------------------------------------------------------------------------------------------------------
----------+
1 row in set (0.057 sec)

d5_96305_sanchit>
```

2. If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
d5_96305_sanchit> create unique index i_unique on orders(onum);
Query OK, 0 rows affected (0.471 sec)
Records: 0  Duplicates: 0  Warnings: 0

d5_96305_sanchit> insert into orders values (3001,20,'1990-10-03',1,2);
ERROR 1062 (23000): Duplicate entry '3001' for key 'orders.i_unique'
d5_96305_sanchit>
```

3. Create an index that would permit salesperson to retrieve his orders

```
d5_96305_sanchit> select * from orders where snum = 1002;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
3 rows in set (0.009 sec)

d5_96305_sanchit> explain analyze select * from orders where snum = 1002;
+-----------------------------------------------------------------------------------------------------------------------------------+
| EXPLAIN                                                                                                                           |
+-----------------------------------------------------------------------------------------------------------------------------------+
| -> Index lookup on orders using i_salesman (snum = 1002)  (cost=0.8 rows=3) (actual time=0.0226..0.033 rows=3 loops=1)
 |
+-----------------------------------------------------------------------------------------------------------------------------------+
1 row in set (0.008 sec)

d5_96305_sanchit>
```

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

```
d5_96305_sanchit> create index i_rating on customers(cnum, rating);
Query OK, 0 rows affected (0.527 sec)
Records: 0  Duplicates: 0  Warnings: 0

d5_96305_sanchit>
```

5. Create an index to speed up searching order on a given date by given customer.

```
d5_96305_sanchit> select * from orders limit 3;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
3 rows in set (0.007 sec)

d5_96305_sanchit> create index i_order on orders(odate,cnum);
Query OK, 0 rows affected (0.605 sec)
Records: 0  Duplicates: 0  Warnings: 0

d5_96305_sanchit> |
```